Introduction to Theoretical Computer Science

CS2490 - Session 13: From REs to Automata, and back!

Pierre Cagne

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September 16, 2026

From REs to NFA

Last time, we constructed a NFA $\mathcal{A}_r$ from any RE r:

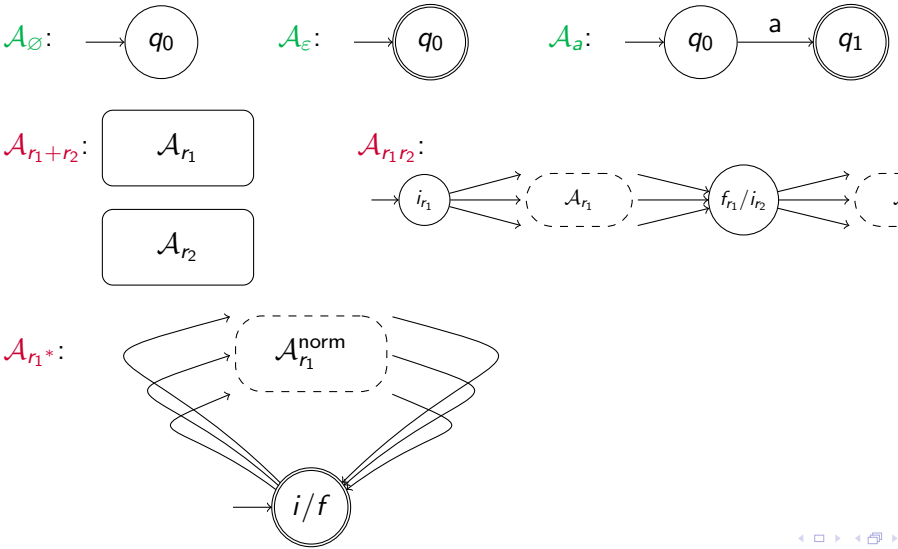
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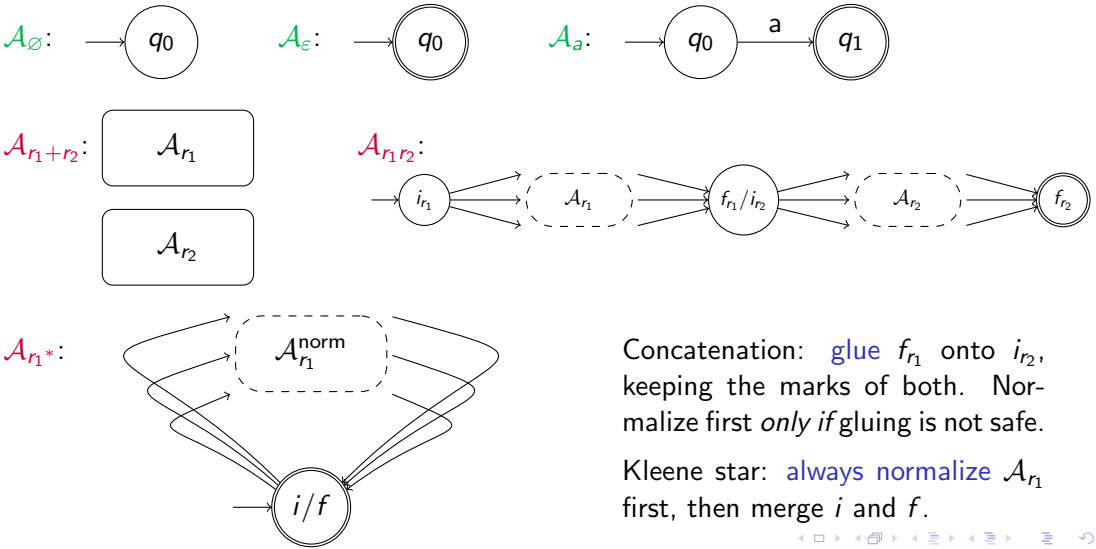
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Concatenation: glue f_{r_1} onto i_{r_2}, keeping the marks of both. Normalize first *only if* gluing is not safe.

Kleene star: always normalize $\mathcal{A}_{r_1}$ first, then merge i and f.

Reminder: when is gluing safe?

Gluing f_{r_1} onto i_{r_2} is safe when:

1. $\mathcal{A}_{r_1}$ has exactly one final state f_{r_1},
2. $\mathcal{A}_{r_2}$ has exactly one initial state i_{r_2},
3. there is no transition out of f_{r_1}, **or** no transition into i_{r_2}.

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If not, first normalize *one* of the two automata: $\mathcal{A}_{r_1}$ if 1. fails, $\mathcal{A}_{r_2}$ if 2. fails, either one if only 3. fails.

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Careful: $\mathcal{A}^{\mathrm{norm}}$ never accepts ε. So after gluing:

- if you normalized $\mathcal{A}_{r_1}$ and $\varepsilon \in \mathrm{L}(r_1)$, make the glue state initial;
- if you normalized $\mathcal{A}_{r_2}$ and $\varepsilon \in \mathrm{L}(r_2)$, make the glue state final.

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Without normalization, this is automatic: the glue state keeps the marks of f_{r_1} and i_{r_2}.

Application

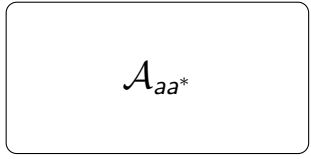
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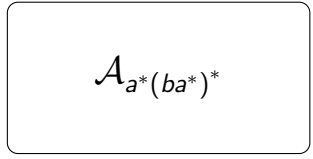
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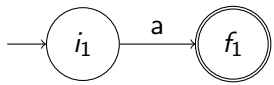


Focus on $\mathcal{A}_{aa^*}$

The RE aa^* is a concatenation, so we first need to construct $\mathcal{A}_a$ and $\mathcal{A}_{a^*}$.

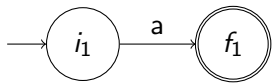
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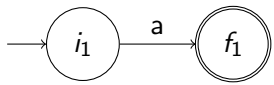
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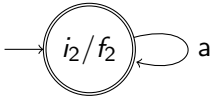
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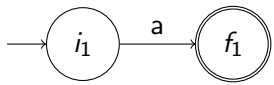
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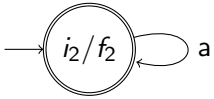
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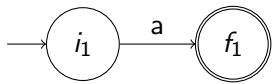


Now concatenate: is gluing f_1 onto i_2/f_2 safe?

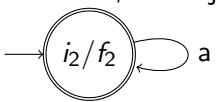
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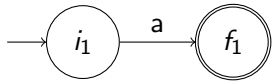
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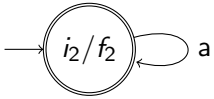
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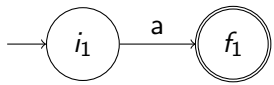
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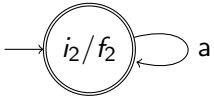
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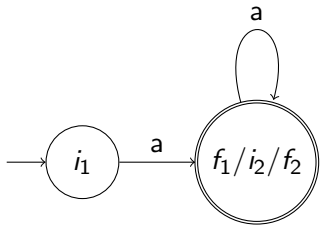
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Focus on $\mathcal{A}_{aa^*}$ (cont'd)

Gluing is safe: no normalization needed.

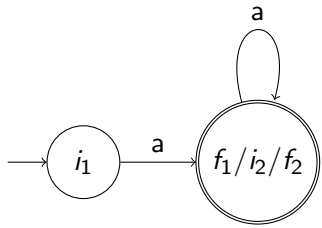
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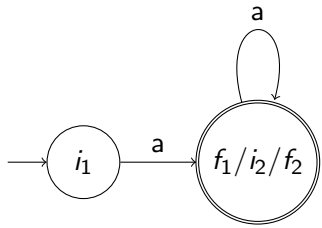
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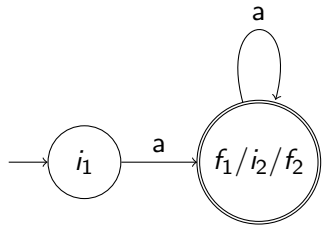
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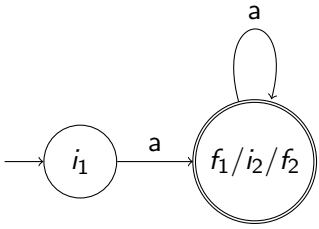
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Compare with the 4-state NFA we would get by normalizing both sides.

Recap so far



$\mathcal{A}_{a^*(ba^*)^*}$

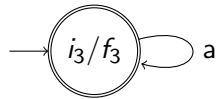
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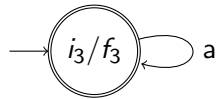
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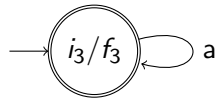


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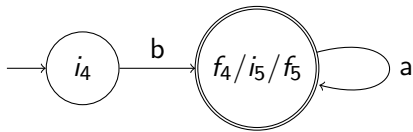
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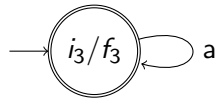
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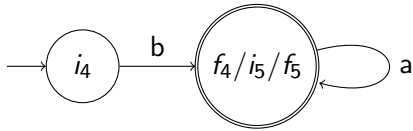
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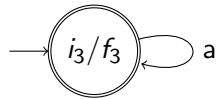


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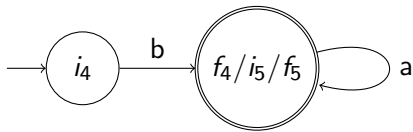
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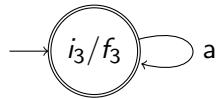


Is $\mathcal{A}_{ba^*}$ normalized? No: there is a transition out of its final state.

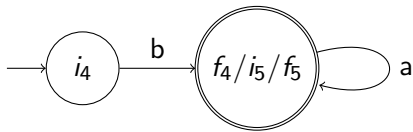
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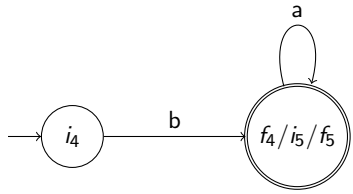


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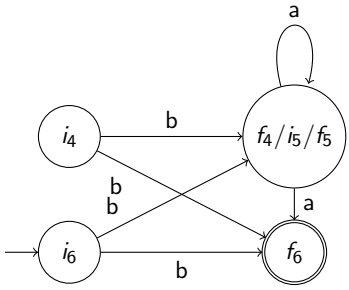
Is $\mathcal{A}_{ba^*}$ normalized? No: there is a transition out of its final state. Normalization is required for the Kleene star.

Focus on $\mathcal{A}_{a^*(ba^*)^*}$ (cont'd)



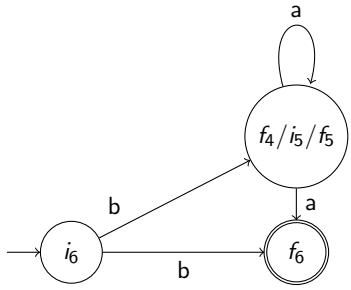
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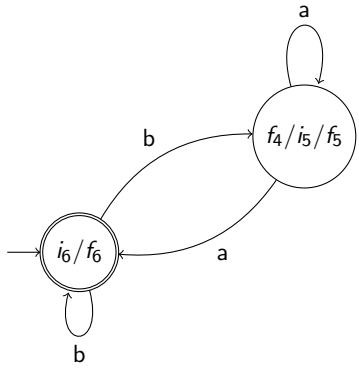
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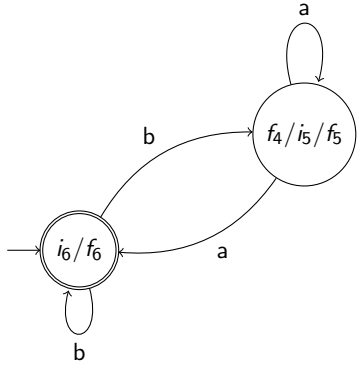
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Focus on $\mathcal{A}_{a^*(ba^*)^*}$ (cont'd)

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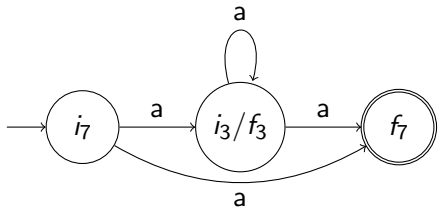
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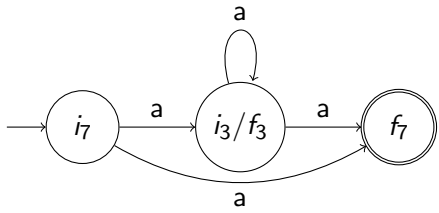


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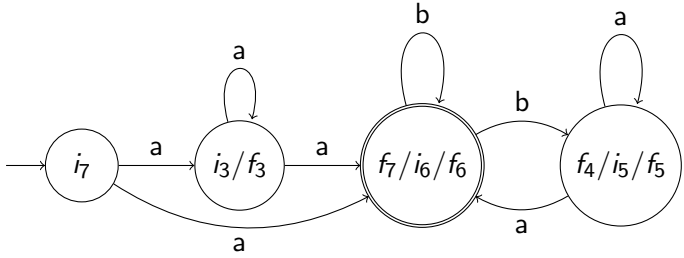
Remember: $\varepsilon \in \mathrm{L}(a^*)$, but $\mathcal{A}_{a^*}^{\mathrm{norm}}$ does not accept it anymore.

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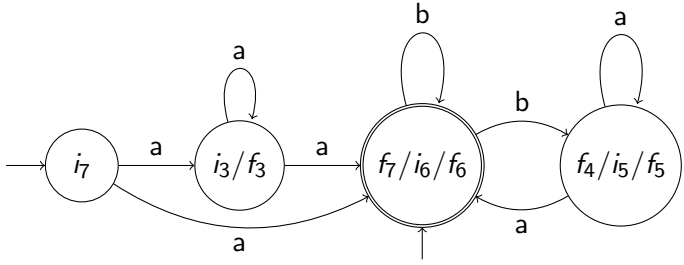
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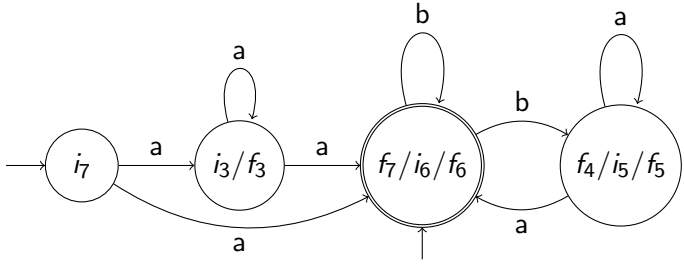
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Sanity check:

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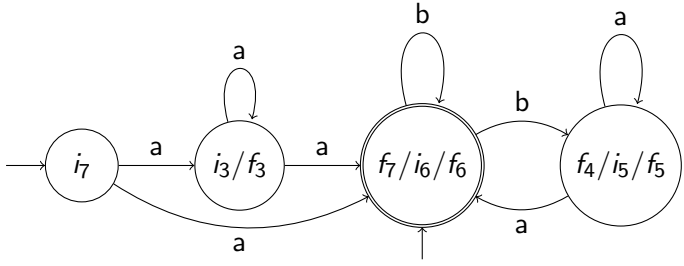
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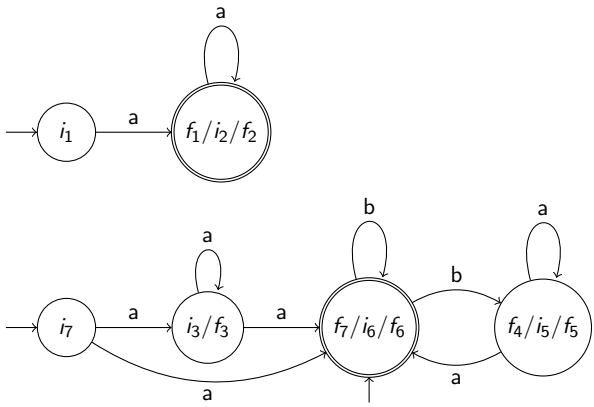
Sanity check: ε and a are accepted (through the glue state), and so are b, ab, ba, $abba$, ...

Recap

The RE $aa^* + a^*(ba^*)^*$ denotes the language recognized by the NFA below:

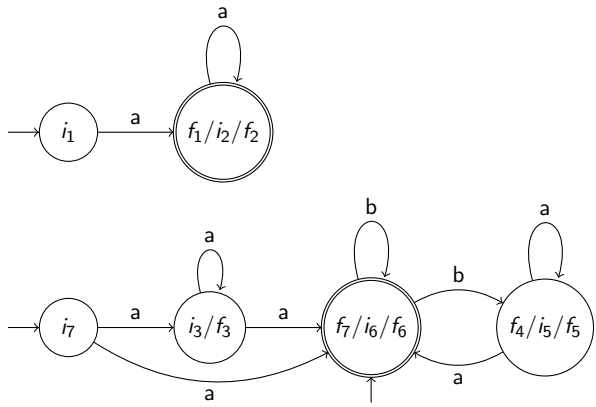
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Compare with the NFA given for L_2 two sessions ago.

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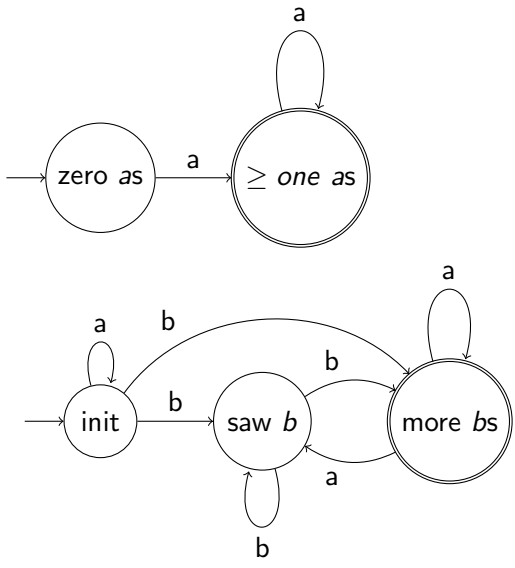
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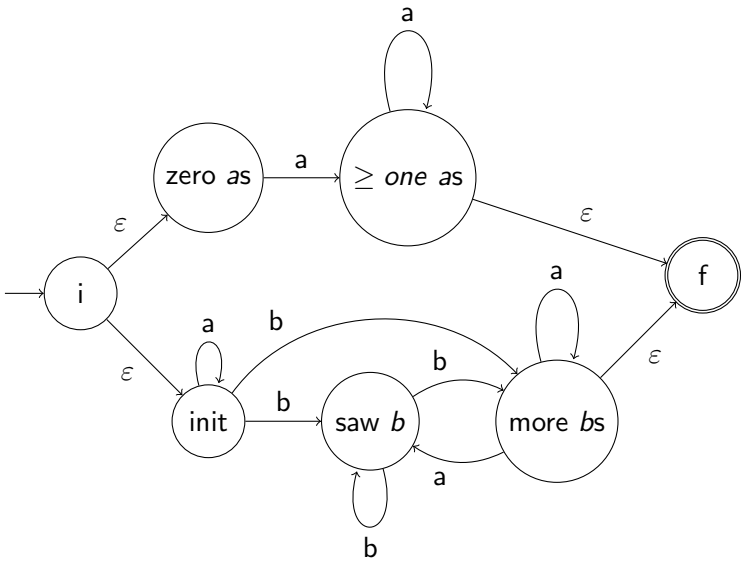
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We stop when we are left with a single initial state i, a single accepting state f, and a transition from i to f labeled with a RE r. Then this r is the regular expression $r_{\mathcal{A}}$ we seek.

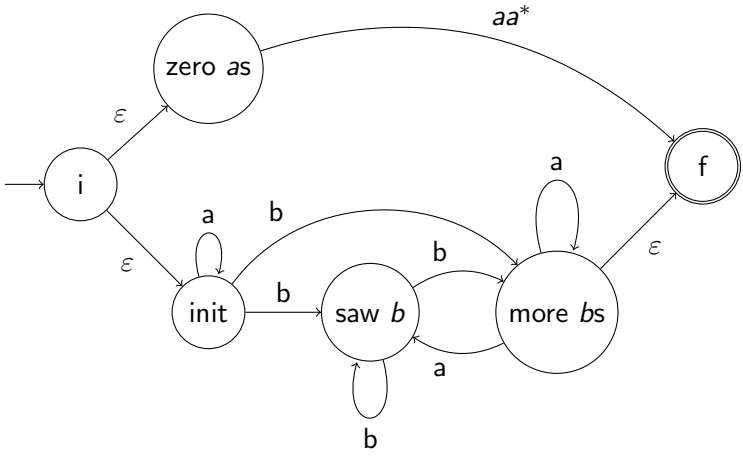
Example: RE for L_2



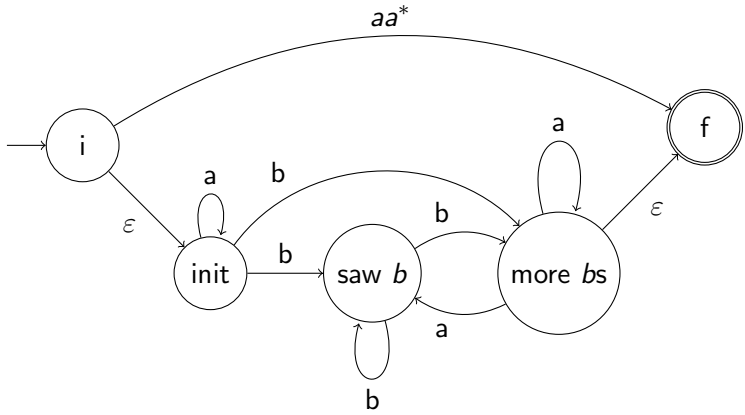
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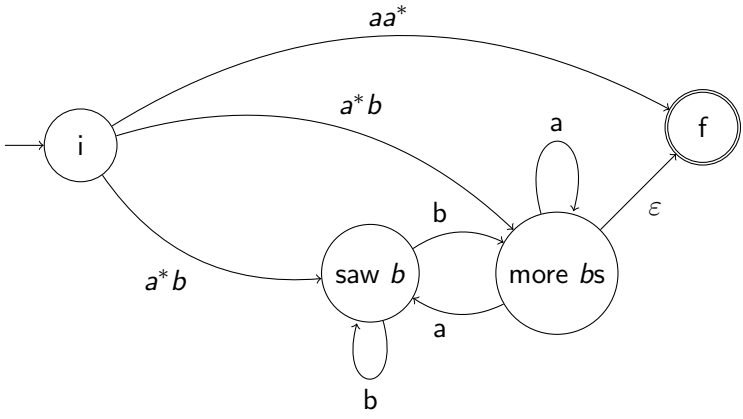
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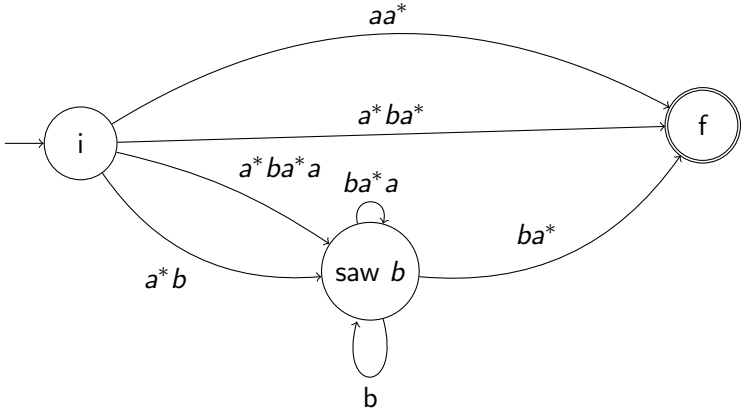
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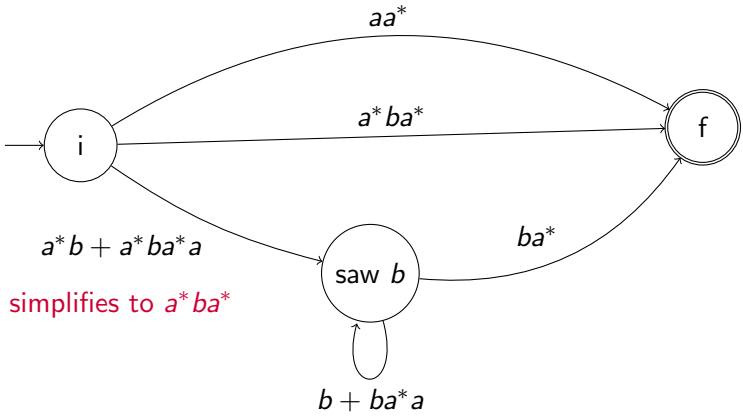
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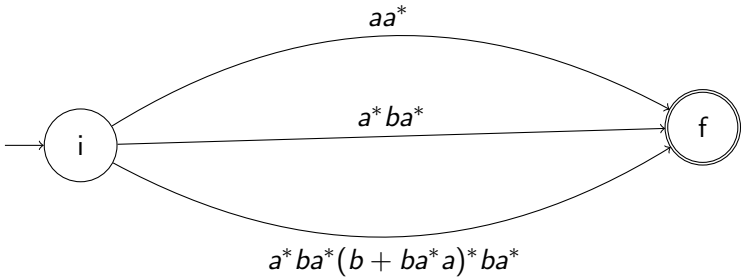
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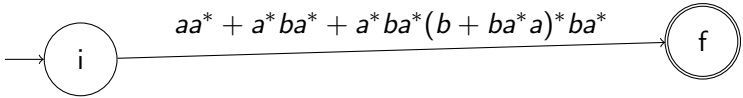
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This is of little importance: these RE are usually generated by a program, and are destined to be read by a machine as well.

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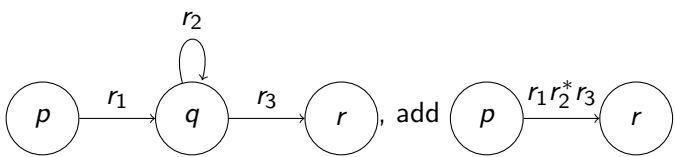
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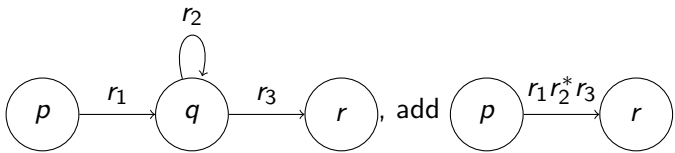
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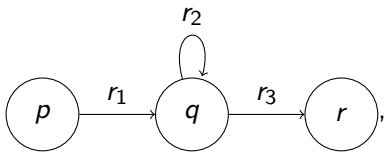
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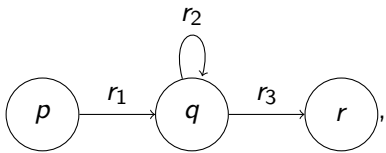
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- For any parallel transitions between two states, combine their labels with $+$.

Practice

Find a RE for the language recognized by the NFA below:

